MANTEY JOSHUA KOJO ADUAMPONG

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EDUCATION

Northeastern University, Boston, MA | Advisor: Dr. Srirupa Chakraborty. | Present
Doctor of Philosophy in Chemical Engineering

Northeastern University, Boston, MA | Sept 2024 – May 2026
Master of Science in Biosystems and Biomaterials

Kwame Nkrumah University of Science and Technology, Ghana | Sept 2019 – Sept 2023
Bachelor of Science in Petrochemical Engineering Class Rank: **5th** /87

TECHNICAL SKILLS

Programming & Data: Python (Scikit-learn, Pandas, NumPy, Matplotlib), MATLAB, MongoDB
Machine Learning & Statistics: MongoDB, Supervised Learning, Model Validation, Cross-Validation, Regression Analysis, Hypothesis Testing, Sensitivity Analysis
Computational Tools: Molecular Dynamics (AMBER, NAMD, Schrödinger, VMD, PyMOL, Chimera), Process Simulation (Aspen HYSYS, ANSYS Fluent), HPC Computing (NSF ACCESS)

EXPERIENCE

Graduate Research Assistant Computational Biology & AI Applications | Jan 2025 – Present
SimBioSys Laboratory, Northeastern University

- Conducted large scale MD simulations of **3+ million** atom MUC2 mucin bile salt systems generating **>10 TB** trajectory data modeling molecular interactions governing interactions through glycosylated mucosal barriers in oral delivery applications.
- Analyzed simulation trajectories using Python (MDAnalysis, NumPy, SciPy) to quantify mucin bile salt interactions enabling data driven formulation design.

Corporate Bankruptcy Risk Model | Financial & Clinical Risk Modeling
Applied Data Science Lab Python • Scikit-learn • Random Forest • Gradient Boosting • Imbalanced Data

- Built ensemble classification models (random forest, gradient boosting) to predict corporate bankruptcy, directly addressing severe class imbalance (~95:5 ratio) through resampling strategies (SMOTE, undersampling).
- Evaluated model performance using precision-recall tradeoffs and F1 metrics appropriate for imbalanced biomedical and financial datasets, with direct transferability to drug failure prediction and clinical risk stratification tasks.

Consumer Segmentation & Interactive Dashboard | US Market Analytics
Applied Data Science Lab Python • Scikit-learn • PCA • k-means • Plotly Dash

- Clustered US consumer population into distinct segments using k-means, applying PCA for dimensionality reduction and visualization of high-dimensional behavioral feature space.
- Built interactive Plotly Dash dashboard to communicate segmentation results, providing a reproducible framework applicable to patient stratification and clinical trial cohort design.

Research Collaborator – HIV Vaccine Development | May 2025 – July 2025
Fred Hutchinson Cancer Center

- Analyzed **8+** conformational states from MD simulations and cryoem structures of HIV envelope proteins using Python to identify conserved binding pockets for broadly neutralizing antibody development.

Air Quality Forecasting System | Nairobi Environmental Monitoring | May 2025 - Jun 2025
Python • MongoDB • Machine Learning • Time Series Analysis

- Engineered end to end ML pipeline processing **200,000+** hourly air quality measurements across Nairobi stored in MongoDB, developing ensemble time series achieving **RMSE < 15 µg/m³** for 24hour PM2.5 forecasts and **82%** accuracy predicting hazardous air quality events.
- Built automated ETL workflow using MongoDB aggregation framework to extract temporal patterns and engineer **50+** features, integrating external datasets and reducing prediction error by **40%** compared to baseline models, enabling data driven public health early warning system for vulnerable population.

Housing Price Prediction System | Mexico City Real Estate Analytics | Feb 2025 - May 2025

- Developed predictive ML model processing **10,000+** real estate listings across Mexico City's Distrito Federal, implementing Ridge regression to forecast apartment prices for data driven real estate investment and market analysis applications.
- Engineered robust preprocessing pipeline handling missing data **15%** of records, extracting hierarchical location features from unstructured address strings, to reduce feature space noise by **40%**.

Computational Modeling Projects | Oct 2024 – May 2025

Northeastern University

- Built ML models analyzing **50,000+** atomic configurations from MD simulations to predict CO_2 adsorption on NH_4HCO_3, informing material optimization for industrial scale carbon capture.
- Processed CFD simulation data (1M+ mesh elements) modeling natural gas flow, heat transfer in underground storage caverns, optimizing injection parameters to enhance storage efficiency and reduce fracture-induced leakage risks.

Bioprocess Engineering : Data Driven Optimization | Sept 2024 – Dec 2024

Northeastern University

- Achieved 250 mg/L recombinant protein yield by analyzing time series bioreactor data across 5+ fermentation runs, applying statistical process control and DOE methods aligned with FDA 21 CFR Part 211
- Implemented PAT framework with real time sensor monitoring to optimize IPTG induction reducing batch variability by 30% through data driven process adjustments.

Science & Engineering Fair (2025) | Member, American Institute of Chemical Engineers

ORGANIZATIONS

- Member, International Association of Engineers (IAENG), Boston, MA
- Member, American Institute of Chemical Engineers, USA.
- Member, Biophysics Society of USA
- Member, African Graduate Students Association, Northeastern University USA
- Member, Ghana Institution of Engineering, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
- Member, Ghana Engineering Student Association
- Assistant Audit Committee Head, Petrochemical Engineering Students Association

JUDGING EXPERIENCE

- Judge, Massachusetts Science and Engineering Fair, Clark University. 05/2025
- Judge, Massachusetts Science and Engineering Fair, Worcester Polytechnic Institute. 04/2025.
- Judge, Sigma Xi Symposium, Undergraduate in Northeastern University. Faculty of Chemical Engineering. 04/2025.

GRANT

- **NSF ACCESS Award** for Computational Research, August 2025
- **Grant Recipient**, New England African Business Expo, Worcester, Massachusetts. May 2025

AWARDS

- **Most Innovative intern,** Lee House & Chemical Ventures, 2022/2023
- **Best Hardworking Employee**, Lee House & Chemical Ventures, 2021/2022
- **Excellent Student Award**, Provost of College of Engineering, KNUST, 2020/2021
- **Excellent Student Award**, Provost of College of Engineering, KNUST, 2019/2020

PRESENTATIONS AND CONFERENCES

- **Mantey Joshua Kojo.A** , Simran Pandey, Rebecca Crrier, Srirupa Chakraborty, "Enhanced Sampling to Study Interactions with Bile Salts", AICHE Annual Meeting-Boston. November 2025
- Atombeat Conference and Learn Series. July 2025
- Festival of Genomics and Biodata. June 2025
- **Mantey Joshua Kojo.A.**, Pandey, S., Chakraborty, S.," Mucosal Interactions with Signaling Molecules", Professional Development, Poster Presentation, Department of Chemical Engineering, Northeastern University. April 2025
- **Mantey Joshua Kojo.A**, "Pyrolysis and Carbonization as a Fuel Preparation Method", Undergraduate Poster presentation, Department of Chemical Engineering, KNUST. March 2025

PUBLICATIONS

- **Mantey Joshua Kojo. A**, Benjamin Edem Meteku, Eric Baffoe, "Amino Acids as Corrosion Inhibitors"- Taylor & Fracis group, 2025.
- Agyei, D., Owusu-Kwarteng, J., Kehinde, B., Ohemeng-Boahen, G., **Mantey Joshua Kojo.A**. "The Role of Indigenous Foods in Achieving Food Security in Africa", 2024.

MEDIA PUBLICATIONS

https://coe.northeastern.edu/news/beyond-the-barrier-how-joshua-mantey-is-reimagining-oral-drug-delivery/
https://www.myjoyonline.com/joshua-mantey-the-use-of-ai-in-drug-delivery/
https://www.myjoyonline.com/digital-intelligence-environmental-disaster-ais-hidden-climate-crisis/